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PEDIATRIC MAJOR HEMORRHAGE

TRANSFUSION

Epsilon aminocaproic acid is associated with acute kidney injury after life-threatening hemorrhage in children

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Abstract

Background: Antifibrinolytic medications have been associated with reduced mortality in pediatric hemorrhage but may contribute to adverse events such as acute kidney injury (AKI).

Study Design and Methods: We conducted a secondary analysis of the MAssive Transfusion in Children (MATIC), a prospectively collected database of children with life-threatening hemorrhage (LTH), and evaluated for risk of adverse events with either antifibrinolytic treatment, epsilon aminocaproic acid (EACA) or tranexamic acid (TXA). The primary outcome was AKI and secondary outcomes were acute respiratory distress syndrome (ARDS) and sepsis.

Results: Of 448 children included, median (interquartile range) age was 7 (2–15) years, 55% were male, and LTH etiology was 46% trauma, 34% operative, and 20% medical. Three hundred and ninety-three patients did not receive an antifibrinolytic (88%); 37 (8%) received TXA and 18 (4%) received EACA. Sixty-seven (17.1%) patients in the no antifibrinolytic group developed AKI, 6 (16.2%) patients in the TXA group, and 9 (50%) patients in the EACA group ($p = .002$). After adjusting for cardiothoracic surgery, cyanotic heart disease, preexisting renal disease, lowest hemoglobin pre-LTH, and total weight-adjusted transfusion volume during the LTH, the EACA group had increased risk of AKI (adjusted odds ratio 3.3 [95% CI: 1.0–10.3]) compared to no antifibrinolytic. TXA was not associated with AKI. Neither antifibrinolytic treatment was associated with ARDS or sepsis.

Conclusion: Administration of EACA during LTH may increase the risk of AKI. Additional studies are needed to compare the risk of AKI between EACA and TXA in pediatric patients.

KEYWORDS

Hemostasis

1 | INTRODUCTION

Hemorrhage is a major cause of death in children.[1,2] The mortality rate of children with life-threatening hemorrhage (LTH) has been measured at 20%–51%, depending on etiology.[2–4] To address the high rate of death from bleeding the concept of damage control resuscitation to include hemostatic resuscitation (balanced blood-based resuscitation) is becoming commonly practiced; however, the use of the lysine analogue antifibrinolytics, tranexamic acid (TXA) and epsilon aminocaproic acid (EACA), in children with LTH is not as common as it is in adult populations.[3,5]

Recent data suggest that TXA increases survival outcomes in adult patients with traumatic injury,[6,7] post-partum hemorrhage,[8,9] non-traumatic intracranial hemorrhage,[10] and all-cause bleeding.[11] Pediatric studies have reported an independent association with improved 24-h survival for patients who receive TXA during an LTH event in observational studies[12,13] and less peri-operative blood loss and need for transfusion in randomized trials compared to placebo.[14–16] Both EACA and TXA have been reported to be associated with decreased morality in children with LTH of all etiologies[12] and blood loss in pediatric cardiothoracic surgery.[17]

In addition to high mortality, adverse events such as acute kidney injury (AKI), acute respiratory distress syndrome (ARDS), and sepsis following an LTH are common.[18] A recent publication in children with LTH from all etiologies, reported a high incidence of AKI of 18.5%, ARDS of 20.3%, and sepsis of 9.8%.[3] However, little data exists to inform if any resuscitative practices are independently associated with increased risk of these outcomes. In an unadjusted analysis of a large multicenter prospective observational study of children with LTH, the MAssive Transfusion in Children (MATIC) study, the use of EACA was associated with risk of AKI, whereas the use of TXA was not.[12] There are currently no data informing the choice of either EACA or TXA as an antifibrinolytic for bleeding children. The decision on which antifibrinolytic to use is therefore dependent on institutional, departmental, and provider preference. In order to improve outcomes following pediatric hemorrhage, more data are needed to identify practices that improve survival while minimizing risk of morbidity. We conducted a secondary analysis of the MATIC database to further investigate the risk of EACA and TXA with adverse events such as AKI, acute respiratory distress syndrome (ARDS), and sepsis.

2 | MATERIALS AND METHODS

2.1 | Study design and data

This is a secondary analysis of the MATIC database, which is comprised of data from 449 children ages 0–17 years with LTH from all causes at 24 tertiary care centers in the United States, Canada, and Italy from January 2014 to October 2018. This analysis was decided post hoc to further evaluate the unadjusted association of EACA with AKI reported in the prior study.[12] LTH was defined as ≥40 mL/kg of total blood products over 6 h or if a patient was transfused after an MTP activation. For patients with multiple bleeding events, only the first event was analyzed. A full description of the MATIC study has been previously published.[3] For this analysis we excluded one patient who received both EACA and TXA during the LTH. The analysis included patients regardless of timing of death in order to reduce survivor bias.

Data analyzed included demographics, etiology of bleeding (trauma, operative, medical), vital signs, laboratory values, injury severity score and/or PRISM III score, Glasgow Coma Scale, and co-morbid conditions (including CNS injury, cardiac arrest, cyanotic heart disease, preexisting renal disease, pre-event vasoactive drug administration, and pre-event use of dialysis, ECMO or mechanical ventilation) prior to the LTH. Data analyzed also included administration of crystalloid before and during LTH, blood product and hemostatic adjunct administration during the event, occurrence of ARDS, AKI, and sepsis in the 7 days following the event, and timing and cause of mortality up to 28 days or hospital discharge. The total dose of EACA and TXA was the total amount during the LTH to include both bolus and continuous infusion dose. The PRISM III score was calculated from data prior to hemorrhage event; missing values were assumed normal per standard calculation.[19] Missing weight variables were assumed at 50th percentile for age based on the Centers for Disease Control and Prevention Clinical Growth Charts.[20] Cause of death was determined by participating sites. The study was conducted under waiver of informed consent granted by each participating center's institutional review board.

2.2 | Analysis

The primary outcome was AKI defined using KDIGO criteria:[21] an increase in serum creatinine ≥0.3 mg/dL (26.5 mmol/L), an increase in serum creatinine ≥1.5 times baseline, or urine volume <0.5 mL/kg/h for 6 hours. Secondary outcomes included ARDS and sepsis. Acute respiratory distress syndrome was defined as a PaO_2/FiO_2 ratio of ≤300 and confirmation of bilateral opacities on chest imaging. Sepsis was defined as a suspected or proven infection caused by any pathogen (by positive culture, tissue stain, or PCR) or a clinical syndrome associated with high probability of infection with evidence on clinical exam, imaging, or laboratory tests.

The occurrence of these outcomes was reported by site coordinators.

For descriptive statistics, patient characteristics were compared between each of the antifibrinolytic groups (TXA, EACA, or none). Continuous data were reported using the mean with standard deviation or the median with the interquartile range. The status of normal distributions was checked using histograms or QQ plots and tested using the Shapiro–Wilk test. Differences in continuous normally distributed data by more than two groups were compared using the analysis of variance (ANOVA) test with a post hoc Bonferroni test. Differences in continuous, non-normally distributed data by more than two groups were tested using the Kruskal–Wallis test and the post hoc nonparametric test of the Dunn test. Differences between any two groups were tested via the *t*-test or Wilcoxon rank sum test as appropriate according to the status of normal distribution. Categorical data were reported as frequencies and percentages and compared using chi-square or Fisher exact test as appropriate.

Univariate and multivariate logistic regression analysis was used to assess factors associated separately with AKI or ARDS in the scope of demographics, bleeding etiology, patient clinical data, resuscitation medications and volumes, and other clinical management. The selection of variables for multivariate analysis was conducted based on prior knowledge of covariate predictors, biological plausibility, the tested hypothesis, and covariates' significance within the data. Candidate variables with a $p < .20$ were included in initial multivariate models. Parameters were sequentially removed manually using backward elimination with a significance cut-off value of $p < .1$ for elimination. The Akaike information criterion (AIC) method was used to estimate the quality of each model in reaching an optimized one.

3 | RESULTS

Of the original 449 children in the MATIC dataset, 448 were included in the sub-analysis. One patient received both TXA and EACA and was excluded. The resulting cohort had a median (interquartile range) age of 7 (2–15) years, a median weight of 25 (12–58) kg, and 55% were male. The etiology of the bleeding was categorized as 46% trauma, 34% operative, and 20% medical. Median PRISM score was 12 (5–22). During the LTH patients received a median total transfusion volume of 61 (29–122) ml/kg over an average duration of 207 (89–367) min. The mortality at 6 h was 15.5%, mortality at 24 h was 22%, and mortality at 28 days was 37%.

The majority of patients (88%) did not receive an antifibrinolytic medication, whereas 8% received TXA and 4% received EACA. The 37 patients who received TXA were located at 17 different sites, the highest number of patients at any site was 11 which represented a mix of operative and trauma bleeding etiologies. The 18 patients who received EACA were located at 12 different sites, the highest number of patients at any one site was 5 which represented a mix of all three bleeding etiologies. Time from initiation of the life-threatening bleeding to TXA administration was median (IQR) = 63 (16–169) min, and a median (IQR) total dose of 21 (14–43) mg/kg of TXA was administered. The dosing regimens when documented were: single bolus ($n = 17$), bolus followed by infusion ($n = 7$), and infusion only ($n = 11$). Time from the initiation of the life-threatening bleeding to EACA administration was median (IQR) = 106 (30–199) min, and a median (IQR) total dose of 110 (54–251) mg/kg of EACA was administered over one ($n = 11$) or two ($n = 7$) doses.

The TXA, EACA, and no antifibrinolytic groups had similar baseline characteristics of sex, etiology of bleeding, pre-existing renal disease and use of dialysis or hemofiltration, pre-LTH mechanical ventilation, continuous vasoactive infusion use, and cardiac arrest (Table 1). However, there were differences in age, extracorporeal membrane oxygenation support prior to hemorrhage, and cyanotic heart disease between the groups. There were also differences among number of patients who received recombinant factor VIIa (highest in the EACA group) and had age-adjusted hypotension (lowest in the TXA group). Patients who received either antifibrinolytic had the longer duration of LTH and received a larger total weight-adjusted transfusion volume as well as weight-adjusted volume of plasma, though the EACA group also received a larger total transfusion volume than the TXA group. Patients in the EACA group were more likely to be younger than the TXA and no-antifibrinolytic groups, have a higher PRISM score, and received a larger weight-adjusted volume of platelets and cryoprecipitate (Table 1).

Regarding the primary outcome of AKI, it occurred in 17.1% of the patients in the no antifibrinolytic group, 16.2% of the patients in the TXA group, and 50% of the patients in the EACA group ($p = .002$) (Table 2). Patients who developed AKI were more likely to have medical or operative bleeding, exposure to EMCO prior to LTH, cyanotic heart disease, exposure to continuous vasoactive infusion at time of event, prior to LTH dialysis and/or pre-existing renal injury, and have received EACA, a larger weight-adjusted transfusion volume of cryoprecipitate, pRBCs, and total blood products (Table 3). Cardiothoracic surgical bleeding was also associated with AKI.

The final adjusted model of best fit accounted for cyanotic heart disease, cardiothoracic surgical bleeding, pre-existing renal disease, pre-LTH hemoglobin, and total

TABLE 1 Characteristics of the treatment groups.

	No antifibrinolytic group 1 ($N = 393$)	Tranexamic acid (TXA) group 2 ($N = 37$)	Epsilon aminocaproic acid (EACA) group 3 ($N = 18$)	p value
Demographics				
Age (years)	7 (2–15)	14 (7–17)	4 (1–12)	.010
Sex (male)	219 (56%)	17 (50%)	11 (61%)	.454
Weight (kg)	25 (11–58)	36 (20–57.8)	20 (8–53)	.273
Clinical data				
Bleeding etiology: (n)				.072
Trauma	184 (47%)	19 (51%)	3 (17%)	
Operative	132 (34%)	10 (27%)	11 (61%)	
Medical	77 (19%)	8 (22%)	4 (22%)	
Mortality in 28 days (n)	148 (38%)	12 (32%)	7 (39%)	.812
Pediatric risk of mortality score	12 (5–22)	12 (5–21)	20 (12–24)	.051
Lowest hemoglobin (g/dl)	9.7 (7.6–11.8)	9.2 (7.4–11.4)	10.2 (8.2–12.2)	.426
Pre-LTH ECMO (n)	22 (6%)	4 (11%)	10 (56%)	<.0001
Continuous vasoactive drug for hemodynamic support	118 (32%)	11 (31%)	9 (50%)	.271
Pre-LTH mechanical ventilation (n)	223 (59%)	18 (53%)	13 (76%)	.257
Pre-LTH cardiac arrest (n)	66 (17%)	6 (17%)	1 (6%)	.512
Pre-existing renal disease (n)	27 (7%)	3 (8%)	2 (12.5%)	.495
Pre-LTH dialysis or CVVHD	15 (4%)	3 (8%)	2 (12.5%)	.079
Cyanotic heart disease (n)	45 (12.2)	0 (0)	8 (47.1)	<.0001
Cardiothoracic surgery (n)	56 (14%)	2 (5%)	9 (50%)	<.0001
Hypotension (n)	232 (59%)	15 (41%)	15 (83%)	.008
Transfusion data				
Duration of LTH (minutes)	196 (77–336)	347 (144–520)	444 (274–600)	.0001
Crystalloid volume prior to and during LTH (ml/kg)	17 (0–51)	32 (8–51)	9 (3–29)	.269
Total blood product during LTH (ml/kg)	58 (28–113)	76 (34–171)	108 (61–216)	.002
RBC during LTH (ml/kg)	30 (15–58)	33 (16–83)	44 (23–104)	.205
Platelet during LTH (ml/kg)	7 (0–18)	8 (4–22)	23 (11–56)	.0002
Plasma during LTH (ml/kg)	16 (4–33)	21 (11–54)	31 (18–130)	.0003
Cryoprecipitate during LTH (ml/kg)	0 (0–2)	0 (0–3)	5 (0–9)	.002
rFVIIa during LTH (yes)	39 (10%)	8 (22%)	8 (44%)	<.0001

Note: Continuous values are given as median (interquartile range) and categorical variables are given as n (%). Hypotension is defined as <3rd percentile for age. The p value is for the comparison of the no antifibrinolytic group, TXA, and EACA groups. For comparisons between two groups for a specific variable the following are the results: Age; Group 1 versus 2, $p = .003$, Group 2 versus 3, $p = .005$. Pre-LTH ECMO; Group 1 versus 3, $p < .001$, Group 2 versus 3, $p = .001$. Cyanotic heart disease; Group 1 versus 2, $p = .023$, Group 1 versus 3, $p = .0001$, Group 2 versus 3, $p < .001$. Cardiothoracic surgery; Group 1 versus 3, $p = .001$, Group 2 versus 3, $p < .001$. Hypotension; Group 1 versus 3, $p = .029$, Group 2 versus 3, $p = .019$. Duration of LTH; Group 1 versus 2, $p = .004$, Group 1 versus 3, $p = .001$. Total blood product during LTH; Group 1 versus 3, $p = .001$. Platelet during LTH, Group 1 versus 3, $p = .001$, Group 2 versus 3, $p = .007$. Plasma during LTH; Group 1 versus 2, $p = .007$, Group 1 versus 3, $p = .012$. Cryoprecipitate during LTH; Group 1 versus 3, $p < .001$, Group 2 versus 3, $p = .003$. rFVIIa during LTH; Group 1 versus 2, $p = .03$, Group 1 versus 3, $p < .001$.
Abbreviations: LTH, life-threatening hemorrhage; rFVIIa, recombinant factor VIIa.

blood product transfusion volume in 10 mL/kg increments (Table 3). In this analysis, children who received EACA as part of their resuscitation had a significantly higher independent risk of developing AKI in the 7 days following the LTH with an odds ratio of 3.26 (95% CI: 1.03–10.33).

TABLE 2 Primary and secondary outcomes by treatment group.

	No antifibrinolytic ($N = 393$)	Tranexamic acid (TXA) ($N = 37$)	Epsilon aminocaproic acid (EACA) ($N = 18$)	*p* value
Acute kidney injury	67 (17%)	6 (16%)	9 (50%)	.002
By bleeding etiology				
Trauma ($n = 206$)	8/184 (4%)	2/19 (11%)	1/3 (33%)	.052
Operative ($n = 153$)	36/132 (27%)	1/10 (10%)	7/11 (64%)	.018
Medical ($n = 89$)	23/77 (30%)	3/8 (38%)	1/4 (25%)	.873
Acute respiratory distress syndrome	78/393 (20%)	5/37 (14%)	7/18 (39%)	.083
Sepsis	38/393 (10%)	4/37 (11%)	2/18 (11%)	.801

TABLE 3 Odds ratios for acute kidney injury from univariate and multivariate logistic regression analysis.

	Univariate odds ratios		Multivariate odds ratios	
	Odds ratio (95% CI)	*p* value	Odds ratio (95% CI)	*p* value
Antifibrinolytic treatment				
No treatment (reference)	1.00		1.00	
TXA	0.94 (0.38–2.35)	.897	0.94 (0.36–2.47)	.905
EACA	4.87 (1.86–12.72)	.001	3.26 (1.03–10.33)	.044
Bleeding etiology				
Trauma (reference)	1.00		-	
Operative	7.16 (3.55–14.43)	.000	-	
Medical	7.72 (3.62–16.46)	.000	-	
Clinical factors				
Age (years)	0.97 (0.94–1.01)	.17	-	
Pediatric Risk of Mortality Score	1.02 (0.99–1.04)	.14	-	
Glasgow Coma Score	1.10 (1.03–1.18)	.006	-	
Pre-LTH hemoglobin	0.92 (0.85–1.00)	.051	0.87 (0.78–0.96)	.005
Duration life-threatening hemorrhage (LTH)	1.0 (1.0–1.0)	.71	-	
Volume pRBCs (ml/kg)	1.003 (1.00–1.005)	.006	-	
Volume plasma (ml/kg)	1.005 (1.00–1.01)	.004	-	
Volume platelets (ml/kg)	1.02 (1.01–1.03)	.000	-	
Total transfusion volume during LTH in 10 mL/kg	1.02 (1.01–1.04)	.001	1.01 (0.996–1.03)	.140
Pre-existing Renal Disease	3.28 (1.55–6.96)	.002	2.78 (1.23–6.29)	.014
Dialysis or hemofiltration	3.10 (1.22–7.84)	.017	-	
Continuous vasoactive drug for hemodynamic support	2.58 (1.57–4.26)	.000	-	
Pre-LTH ECMO	4.19 (2.06–8.51)	.000	-	
Cyanotic heart disease	5.02 (2.73–9.24)	.000	2.62 (1.09–6.29)	.031
Cardiothoracic surgery bleeding	4.35 (2.47–7.64)	.000	2.46 (1.06–5.70)	.035

Note: The alternative covariates that were considered in model generation included age, bleeding etiology, PRISM score, pre-LTH ECMO, age-adjusted hypotension, receipt of platelets, continuous vasoactive drug infusion, hemodialysis, volume of RBCs, GCS score, and total volume of plasma.
Abbreviations: EACA, epsilon aminocaproic acid; LTH, life-threatening hemorrhage; TXA, tranexamic acid.

Secondary outcomes of ARDS and sepsis were evaluated (Table 2). Neither were significantly associated with either treatment group on univariate analysis. ARDS met the threshold for further evaluation with multivariate analysis; however, the multivariate model of best fit did not reveal a significant association between the treatment group and ARDS.

4 | DISCUSSION

In this secondary analysis of an international, multicenter prospective study in children with LTH, EACA was independently associated with increased risk of AKI while TXA was not. This is the first study to compare risk of AKI associated with the lysine analogue antifibrinolytics in pediatric patients with multiple etiologies of LTH. There is currently no high-quality data informing the choice of antifibrinolytic. Both medications have been associated with decreased mortality and less blood loss in certain populations; thus, it is helpful to identify risks associated with these antifibrinolytics to guide resuscitation practices.

Early concerns about EACA included renal injury from intrarenal obstruction due to intrarenal coagulation and fibrin deposition, postulated to occur secondary to its excretion as unchanged drug in the urine (pharmacokinetics shared by TXA) with several early case reports purporting this complication.[22,23] However, EACA has not been associated with renal dysfunction in multiple recent studies in non-LTH cohorts, including a recent large observational study in adult cardiopulmonary bypass patients and a 2011 Cochrane review of perioperative antifibrinolytic use.[24,25]

One theoretical explanation for the worsening of renal function after administration of an antifibrinolytic is extreme suppression of fibrinolytic activity, which has been termed fibrinolysis shutdown and has been studied most extensively in trauma patients. Fibrinolysis shutdown has been defined by the inflection point of increased mortality by thromboelastographic analysis of LY30 < 0.9% (physiologic fibrinolysis LY30 = 0.9%–2.9%; hyperfibrinolysis LY30 ≥ 3.0%).[26] Fibrinolysis shutdown has been associated with increased mortality in adult trauma patients[27,28] and mortality and deep venous thrombosis in pediatric trauma patients.[29] Meizoso et al. found an association of TXA administration during trauma resuscitation with fibrinolysis shutdown.[30] Interestingly, they report an increased incidence of AKI and acute lung injury in the patients with fibrinolysis shutdown regardless of TXA administration and no association with TXA administration itself with AKI. As the MATIC data predates widespread use of thromboelastographic analysis at children's hospitals, we did not have sufficient data to evaluate whether fibrinolysis shutdown is more common in the patients who received EACA or those who developed AKI. This offers a theoretical explanation for our findings of increased rates of AKI in those who received EACA that should be explored in future studies.

An imbalance in equivalent dosing could explain either increased fibrinolysis shutdown or drug-induced renal injury in the EACA group. Patients who received EACA received a median total dose of 110 (IQR: 54–251) mg/kg, while those who received TXA received a median total dose of 21 (14–43) mg/kg. Standard dosing of the antifibrinolytics for use in pediatric hemorrhage is not well-established. The ideal therapeutic serum concentration of TXA for bleeding children is not confirmed by large in vitro *studies*, though a systematic review of pediatric and adult in vivo and in vitro studies suggests 10–15 μg/mL.[31] Pharmacokinetic studies in pediatrics have targeted concentrations of 20–150 μg/mL. These studies have recommended dosing schemes of a bolus dose of 4–120 mg/kg followed by a continuous infusion of 2–17 mg/kg/hr based on the provider's desired target concentration, patient weight, and exposure to cardiopulmonary bypass.[32–34] The therapeutic serum concentration of EACA has been considered to be 130 μg/mL based on in vitro studies of inhibition of bovine plasminogen in the 1950s and was the target of subsequent suggested dosing regimens in children.[35–37] More recent in vitro studies using adult and pediatric plasma have shown fibrinolysis is inhibited at much smaller concentrations of EACA, with one neonatal study finding the effective concentration to be about 40 μg/mL concentration.[35,38] Thus, commonly accepted pediatric dosing regimens for EACA could be targeting supratherapeutic concentrations, increasing the risk of AKI.

While our results showed a significant association of EACA with AKI in operative patients and approached significance in trauma patients; there was no association in patients with medical bleeding. This may be explained by the heterogeneity in bleeding etiology among medical patients, smaller sample size, and the fact that medical etiology AKI risk by itself exceeded EACA-associated risk in our population. Our findings are reflective of the broader literature available for medical bleeding in which the strategies that show survival benefit in trauma and surgical bleeding had limited reported benefit.[39–42] As the bleeding risks and coagulation profiles of medical patients are likely to be significantly different from operative and trauma patients, it is also possible that antifibrinolytic medications have different physiologic effects in this heterogenous population. Large, multi-center studies in children with medical bleeding are needed to determine the efficacy and risk profile of antifibrinolytics in this population.

A limitation of this study is that it was observational in nature as well as a secondary analysis of a dataset, thus, analysis was limited to existing data and underpowered for some comparisons. While we attempted to account for confounding variables in our analysis, there could be other risk factors that offer an explanation of these results, especially given the selection of antifibrinolytics is often based on subspecialty preference. For example, EACA is frequently used in the cardiothoracic surgery population which may have been exposed to cardiopulmonary bypass, a risk factor for AKI. While exposure to bypass was not specifically reported in this dataset, our model of best fit adjusted for perioperative cardiothoracic surgery bleeding and cyanotic heart disease, which likely captured the risk of cardiopulmonary bypass as a confounding variable. As previously discussed, functional coagulation assays were not commonly used at the time of the data collection, thus we could not make a full assessment of each group's global hemostatic function to include fibrinolytic activity. A last limitation of this study is inability to report number of thrombotic events and seizures, other potentially important adverse outcomes associated with antifibrinolytic therapy, as this information was not collected. However, reported rates of these complications in pediatric patients receiving anti-fibrinolytics are low.[43] Given the limitations of this observational study, we intend these results to be hypothesis-generating rather than lead to changes in clinical practice on their own. Large, multicenter randomized trials are needed to further assess these results.

5 | CONCLUSIONS

EACA administered during an LTH event was independently associated with AKI within 7 days, while TXA was not. Optimal dosing of EACA needs further evaluation before randomized controlled trials are performed to evaluate the indication and outcomes for the use of either EACA or TXA for LTH in children.

CONFLICT OF INTEREST STATEMENT

The authors have disclosed no conflicts of interest.

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